

DOCUMENTACIÓN TÉCNICA

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Matricula	2025-0773
Asignatura	Seguridad en Redes
Script	AshleyFabian_2025-0773_DHCP_Starvation_P1.py
Practica	P1
Fecha	05 de June de 2026
Herramienta	Python 3 + Scapy
Plataforma	GNS3 — Kali Linux

1. Objetivo del laboratorio

Demostrar el agotamiento del pool DHCP del router CSR1000v enviando 254 DHCPDISCOVER con MACs aleatorias desde Kali Linux (192.168.73.10). El pool VLAN10 tenia 254 IPs disponibles (192.168.73.1-254, excluyendo .1-.9). Tras el ataque, 244 de 254 IPs quedaron en estado Selecting, impidiendo que nuevos hosts obtengan IP.

- Agotar el pool DHCP VLAN10 del router CSR1000v.
- Verificar con `show ip dhcp binding` que la tabla esta llena de MACs falsas.
- Confirmar que la VPCS no puede obtener IP tras el ataque.
- Aplicar rate-limiting de DHCP Snooping como contra-medida.

2. Objetivo del script

El script `AshleyFabian_2025-0773_DHCP_Starvation_P1.py` genera DHCPDISCOVER con MACs aleatorias en el campo chaddr del BOOTP header. Cada MAC unica es interpretada por el servidor como un cliente diferente que reserva una IP del pool. Se enviaron 254 paquetes agotando casi todo el pool.

2.1 Parametros

Parametro	Descripcion	Valor usado
-i / --iface	Interfaz de red	eth0
-c / --count	Cantidad de DISCOVER (0=infinito)	254
-d / --delay	Delay entre paquetes (segundos)	0

2.2 Requisitos

- Kali Linux con Python 3 y Scapy.
- Servidor DHCP activo en CSR1000v (pool VLAN10: 192.168.73.0/24).
- Kali en VLAN 10 conectada a SW1 Gi0/1.
- Ejecutar como root.

2.3 Uso

```
sudo python3 AshleyFabian_2025-0773_DHCP_Starvation_P1.py -i eth0 -c 254

# Verificar en router
Router# show ip dhcp binding
Router# show ip dhcp pool
```

3. Funcionamiento del script

- Genera MAC aleatoria con prefijo 02: por iteracion.

- Construye DHCPDISCOVER: Ethernet broadcast -> IP 0.0.0.0->255.255.255.255 -> UDP 68->67 -> BOOTP con chaddr=MAC_falsa -> DHCP discover.
- Envía con sendp() en capa 2.
- Repite 254 veces sin delay para maxima velocidad.

3.1 Resultados obtenidos

Metrica	Antes del ataque	Despues del ataque
Leased addresses VLAN10	0 de 254	244 de 254
Estado de las entradas	—	Selecting (sin completar handshake)
IPs disponibles	245 (exc. 9)	10 restantes
Deteccion del router	—	PING_CONFLICT en 192.168.73.10
VPCS puede obtener IP	Si	No (pool agotado)

4. Documentacion de la red

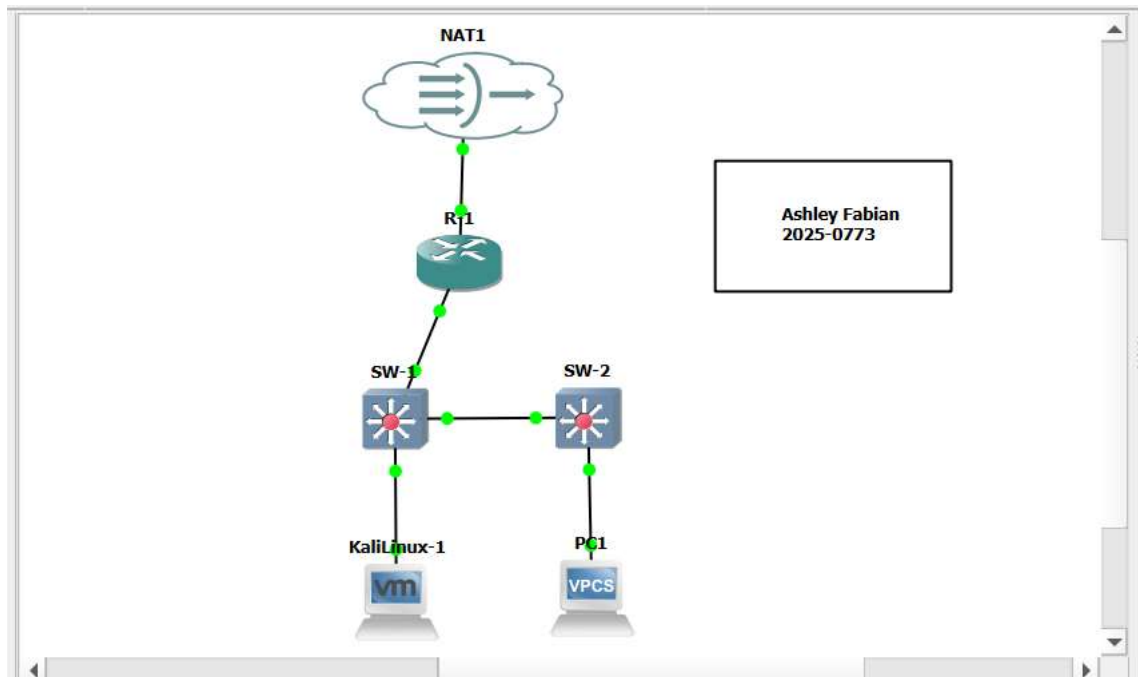
Dispositivo	Rol	Interfaz / Puerto	IP / VLAN
CSR1000v	Router / Gateway	Gi2.10 — Gi2.20	Gi2.10: 192.168.73.1/24 Gi2.20: 192.168.74.1/24
SW1 — vIOS L2	Switch capa 2	Gi0/0 trunk R1 Gi0/1 VLAN10 Kali Gi0/2 trunk SW2	VLANs 10 (ATACANTE) y 20 (VICTIMA)
SW2 — vIOS L2	Switch capa 2	Gi0/0 VLAN20 VPCS Gi0/1 trunk SW1	VLANs 10 (ATACANTE) y 20 (VICTIMA)
Kali Linux	Atacante	eth0	192.168.73.10/24 — VLAN 10
VPCS	Victima	eth0	192.168.74.20/24 — VLAN 20

4.1 Configuracion DHCP en CSR1000v

```
ip dhcp excluded-address 192.168.73.1 192.168.73.9
ip dhcp pool VLAN10
network 192.168.73.0 255.255.255.0
default-router 192.168.73.1
dns-server 8.8.8.8
!
ip dhcp excluded-address 192.168.74.1 192.168.74.9
ip dhcp pool VLAN20
network 192.168.74.0 255.255.255.0
default-router 192.168.74.1
dns-server 8.8.8.8
```

5. Capturas de pantalla

Topología con nombre y matrícula visibles:



Muestra del router antes del ataque:

The screenshot shows a Kali Linux desktop with a terminal window open. The terminal output is as follows:

```
Router-AF-1025-0773#show ip dhcp pool

Pool: VLAN20
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses               : 0
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index   IP address range   Leased/Excluded/Total
  192.168.73.1    192.168.73.1 - 192.168.73.254 0 / 0 / 254

Pool: VLAN20
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses               : 0
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index   IP address range   Leased/Excluded/Total
  192.168.74.1    192.168.74.1 - 192.168.74.254 0 / 0 / 254

Router-AF-1025-0773#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration   Type   State   Interface
Hardware address/
User name

Router-AF-1025-0773#
```

[illegible]

The screenshot shows a Kali Linux virtual machine environment. The terminal window displays the configuration of a DHCP pool named 'pool1' on the 'vswan0' interface. The configuration includes the pool name, the interface, and the pool range (192.168.73.1 to 192.168.73.254). The DHCP pool is currently in the 'selecting' state. The network configuration file /etc/network/interfaces shows the vswan0 interface configured with a static IP address of 192.168.73.1 and a netmask of 255.255.255.0. The DHCP pool is also configured to use the 'vswan0' interface. The DHCP pool is currently in the 'selecting' state.

Muestra de cómo la VPCS falla al querer obtener una IP por DHCP:



```
PC1> no ip 192.168.73.20 255.255.255.0 192.168.73.1
Bad command: "no ip 192.168.73.20 255.255.255.0 192.168.73.1". Use ? for help.

PC1> ip 192.168.73.20 255.255.255.0 192.168.73.1
Checking for duplicate address...
PC1 : 192.168.73.20 255.255.255.0 gateway 192.168.73.1

PC1> ip 192.168.73.20 255.255.255.0 192.168.73.1
Checking for duplicate address...
PC1 : 192.168.73.20 255.255.255.0 gateway 192.168.73.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.73.20/24
GATEWAY    : 192.168.73.1
OHS       :
MAC        : 00:50:79:66:68:00
LPORT     : 10000
RHOST:PORT : 127.0.0.1:10001
MTU        : 1500

PC1> ping 192.168.73.1
host (192.168.73.1) not reachable

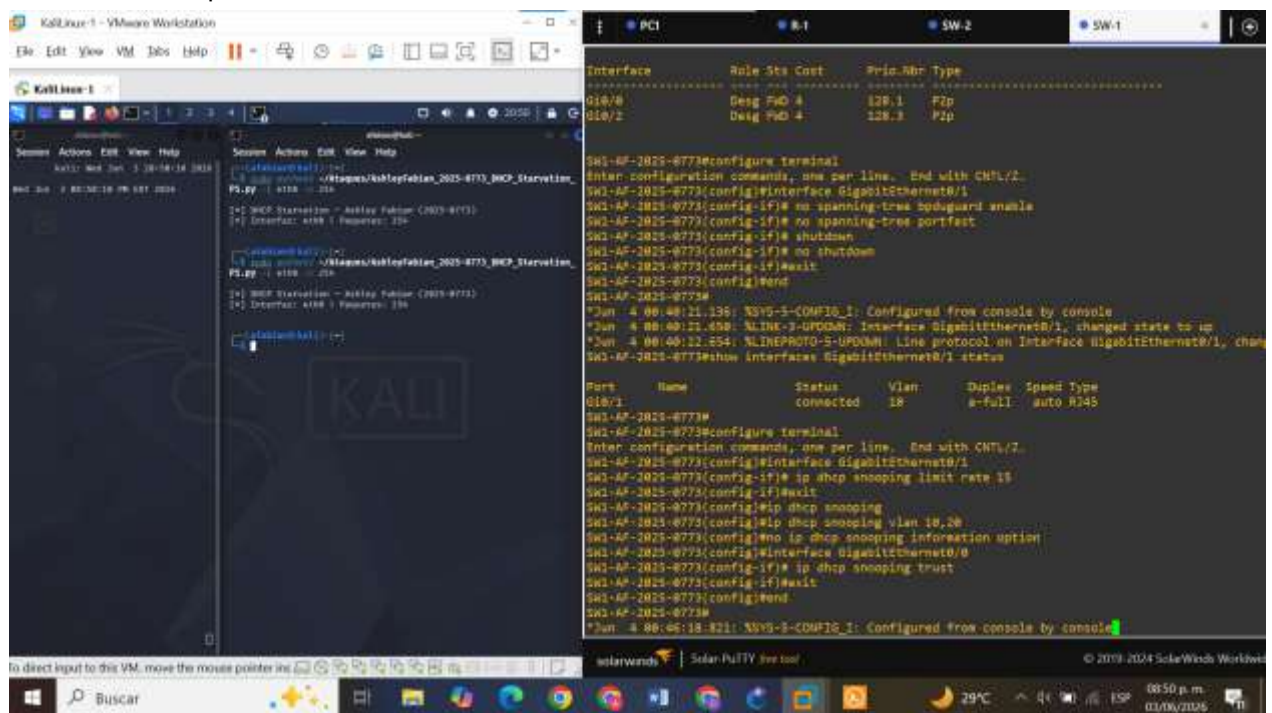
PC1> ping 192.168.73.10
host (192.168.73.10) not reachable

PC1>
PC1> ip 192.168.74.20 255.255.255.0 192.168.74.1
Checking for duplicate address...
PC1 : 192.168.74.20 255.255.255.0 gateway 192.168.74.1

PC1> dhcp
DDO
Can't find dhcp server

PC1>
```

Contra-medida aplicada:



```
Interface      Rule Sta Cost      Prio Rbr Type
-----
Gig0/0        Desig Prio 4      128.1 P2p
Gig0/2        Desig Prio 4      128.3 P2p

SW1-AF-2025-0773#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-AF-2025-0773(config)#interface GigabitEthernet0/1
SW1-AF-2025-0773(config-if)# no spanning-tree bpduguard enable
SW1-AF-2025-0773(config-if)# no spanning-tree portfast
SW1-AF-2025-0773(config-if)# shutdown
SW1-AF-2025-0773(config-if)# no shutdown
SW1-AF-2025-0773(config-if)#exit
SW1-AF-2025-0773(config)#end
SW1-AF-2025-0773#

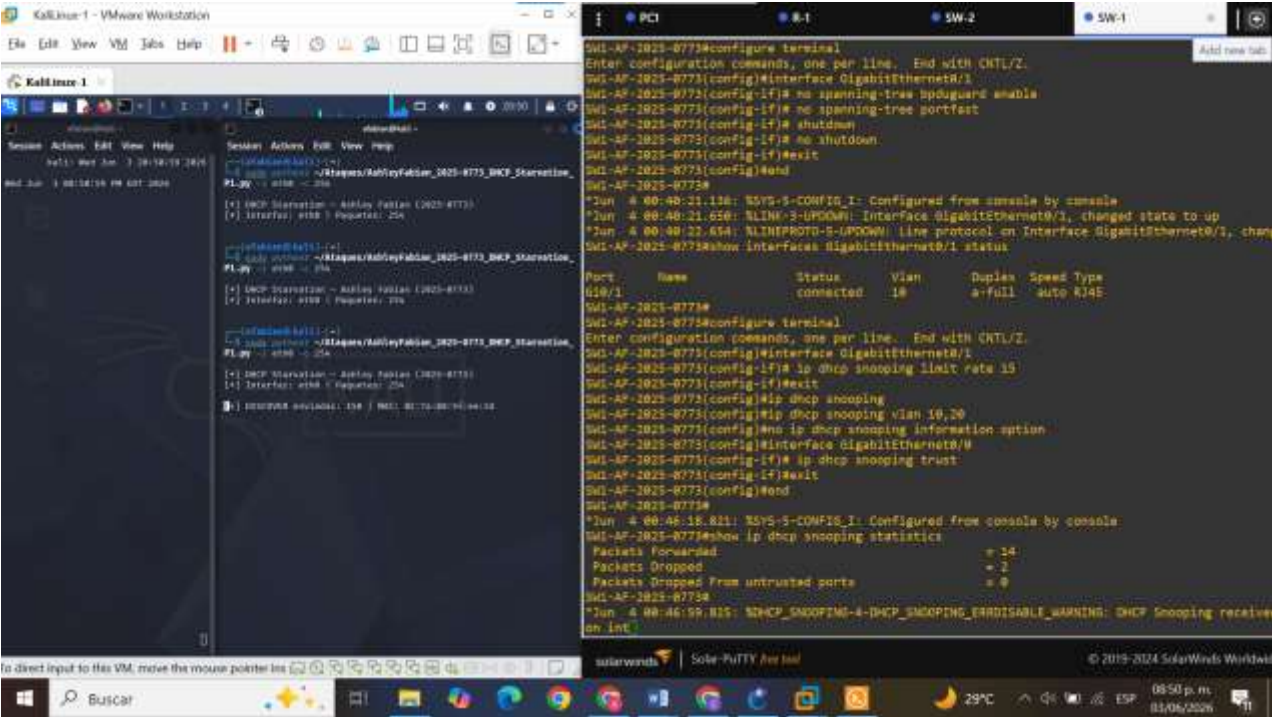
*Jun  4 00:40:21.135: XSV5-S-COMP10_1: Configured from console by console
*Jun  4 00:40:21.450: XLINE-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun  4 00:40:22.554: XLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW1-AF-2025-0773#show interfaces GigabitEthernet0/1 status

Port      Name      Status      Vlan      Duplex  Speed Type
-----
Gig0/1    Gig0/1    connected   10        a-FULL  auto  RJ45

SW1-AF-2025-0773#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-AF-2025-0773(config)#interface GigabitEthernet0/1
SW1-AF-2025-0773(config-if)# ip dhcp snooping limit rate 15
SW1-AF-2025-0773(config-if)#exit
SW1-AF-2025-0773(config)#ip dhcp snooping
SW1-AF-2025-0773(config)#ip dhcp snooping vlan 10,20
SW1-AF-2025-0773(config)#no ip dhcp snooping information option
SW1-AF-2025-0773(config)#interface GigabitEthernet0/0
SW1-AF-2025-0773(config-if)# ip dhcp snooping trust
SW1-AF-2025-0773(config-if)#exit
SW1-AF-2025-0773(config)#end
SW1-AF-2025-0773#

*Jun  4 00:40:19.821: XSV5-S-COMP10_1: Configured from console by console
```

Resultado de la contra-medida aplicada:



6. Contra-medidas

Contra-medida	Comandos aplicados	Resultado
DHCP Snooping Rate-limit	ip dhcp snooping ip dhcp snooping vlan 10 (Gi0/1)# ip dhcp snooping limit rate 15	Bloquea el puerto si supera 15 DHCP pps
Port Security	switchport port-security maximum 5 switchport port-security violation restrict	Limita MACs unicas por puerto